FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Advanced)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CRT– I
(Paper-1)

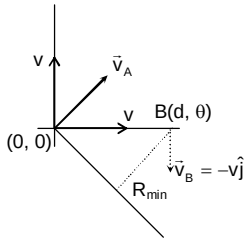
Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A, B**	**D**	**A, B, C, D**
2.	**C, D**	**A, B, C**	**A, B, C**
3.	**A, B, D**	**A, C**	**C, D**
4.	**A, C**	**B, C**	**A, B, C, D**
5.	**A, B**	**A, C**	**A, B**
6.	**A, D**	**B**	**A, B, C, D**
7.	**C, D**	**B**	**B, D**
8.	**A, C**	**A, D**	**A, D**
9.	**A, D**	**B, D**	**A, B, C**
10.	**B, D**	**A, B**	**A, D**
1.	**2**	**3**	**0**
2.	**0**	**3**	**1**
3.	**2**	**4**	**3**
4.	**3**	**3**	**3**
5.	**4**	**7**	**1**
6.	**3**	**8**	**1**
7.	**2**	**9**	**4**
8.	**5**	**3**	**2**
9.	**0**	**0**	**4**
10.	**3**	**2**	**2**

Physics

PART – I

SECTION – A

1. Using concept of relative velocity we can solve



3. Use $v^2 = u^2 + 2as$
And impulse = change in momentum

4. Use $v = \sqrt{\frac{2gh}{\beta}}$

where, $\beta = 1 + \frac{I}{Mr^2}$

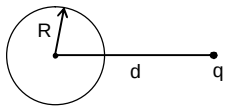
5. Use work energy theorem

6. Buoyant force $F_B = V\rho_e g_{eff}$
If lift accelerates
$g_{eff} = g + a$

7. Use, $\nu_{app} = \nu_v \left(\frac{v - v_0}{v - v_s} \right)$
v = speed of sound

8. for isothermal process, PV = constant
For adiabatic process, PV^γ = constant

9. Electric field at centre = 0
So, charge q will produce same electric field as induced charge on sphere at centre so A & D



10. Using conservation linear moment and conservation mechanical energy.

SECTION – C

1. For translation position of A will be 2.

2. formula for rod and are both are
$\frac{k\lambda}{d}(\sin\alpha + \sin\beta)\hat{i} + \frac{k\lambda}{d}(\cos\alpha - \cos\beta)\hat{j}$
So, $\vec{E} = \vec{0}$ and b = 0

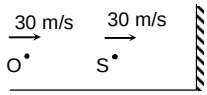
6. work done = $\frac{3}{2}P_0V_0$

$\Delta U = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$

$\Delta Q = \frac{3}{2}P_0V_0$

$\therefore$ k = 3

7. $f_1 = f$



$f_w = \frac{C}{C-30}f = \frac{330}{300}\times 1000$ Hz

$f_2 = \frac{C+v_0}{C}f_w = \frac{360}{330}\times\frac{330}{300}\times 1000 = 1200$ Hz

Frequency Band width = $f_2 - f_1$ = 200 Hz

8. $F_{net} = F_{pseudo} - F_{Buoyant} = V\times 2\rho\times 10 - v\rho\times 10 = v\rho\times 10$

$a = \frac{F_{net}}{n} = \frac{v\rho\times 10}{v\times 2\rho} = 5$ m/s

9. $I_L = \frac{100}{50\sqrt{2}} = \sqrt{2}$ A

$I_C = \frac{100}{50\sqrt{2}} = \sqrt{2}$

$I_{source} = I_L - I_C = 0$

Chemistry

PART – II

SECTION – A

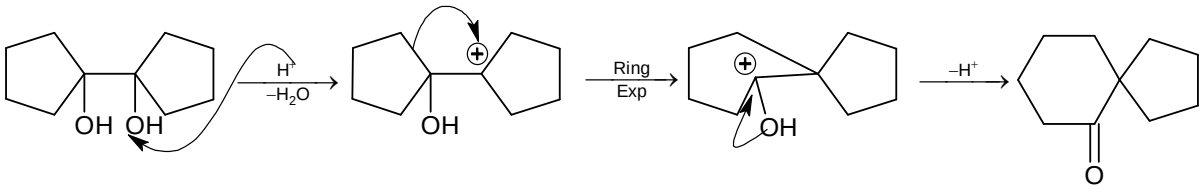
1. Dacron is a copolymer of ethylene glycol and terephthalic acid

2. $${}^{A}_{Z}A \xrightarrow{-\alpha} {}^{A-4}_{Z-2}B \xrightarrow{-\beta} {}^{A-4}_{Z-1}C \xrightarrow{-\beta} {}^{A-4}_{Z}D$$
α - emission gives isodiapher.

3. From kinetic theory of gases
K. E. $\propto$ T

4. $$M_{Na_2CO_3} = \frac{W_B \times 1000}{m_B \times V} = \frac{5.3 \times 1000}{106 \times 100} = 0.5 \qquad M_{H_2C_2O_4.2H_2O} = \frac{W_B \times 1000}{m_B \times V} = \frac{6.3 \times 1000}{126 \times 100} = 0.5$$
$$N_{Na_2CO_3} = \frac{W_B \times 1000}{E_B \times V} = \frac{5.3 \times 1000}{53 \times 100} = 1 \qquad N_{H_2C_2O_4.2H_2O} = \frac{N_B \times 1000}{E_B \times V} = \frac{6.3 \times 1000}{63 \times 100} = 1$$
n-factor =2 n-faction = 2

5. $E^o_{\text{redox change}}$ will be negative for non-spontaneous reaction
$$Ni^{2+}(aq) + Cu(s) \longrightarrow Ni(s) + Cu^{2+}(aq)$$
$$E^o = E^o_{Ni^{2+}/Ni} - E^o_{Cu^{2+}/Cu} = -0.25 - 0.34 = -0.59V\,(-ve,\ non-spontaneous)$$
$$Cu(s) + 2H^+(aq) \longrightarrow Cu^{2+}(aq) + H_2(g)$$
$$E^o = E^o_{H^+/H_2} - E^o_{Cu^{2+}/Cu} = 0 - 0.34 = -0.34V\,(-ve,\ non-spontaneous)$$

6. 

7. $$CH_3CHO \xrightarrow[(ii)\ H_2O]{(i)\ CH_3MgBr} H_3C-\underset{\underset{OH}{|}}{\overset{\overset{H}{|}}{C}}-CH_3 \xrightarrow{H_2SO_4/\Delta} \underset{(B)}{CH_3-CH=CH_2} \xrightarrow{HBO} \underset{(C)}{CH_3-CH_2-CH_2-OH}$$
(B)

8. Compounds have all 'c' atom sp^2 hybridised are coplanar.

9. B. O. of NO = 2.5, paramagnetic
NO^+ = 3.0, diamagnetic

10. Cr^{3+}, d^3 configuration shows same number paired e^- and EAN but strength of all ligands are different so does not slow same colour and splitting.

SECTION –C

1. [Mabcd] type square planar complex has 3 G. I.

2. $NO_2^- + HCl \longrightarrow HNO_2 + Cl^-$

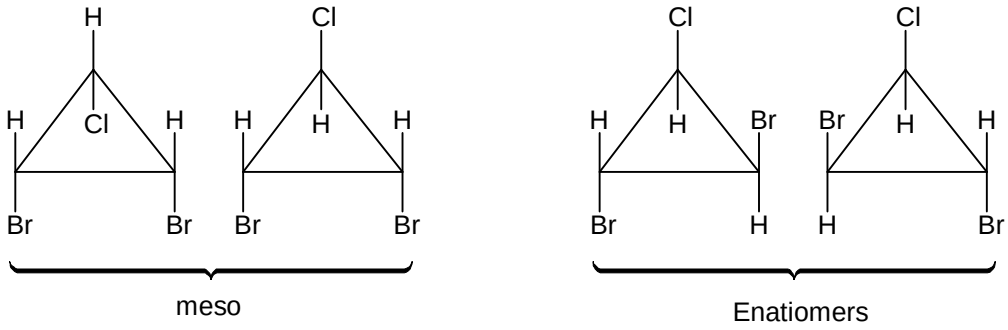
$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$

$2NO + O_2 \longrightarrow 2NO_2 \uparrow$

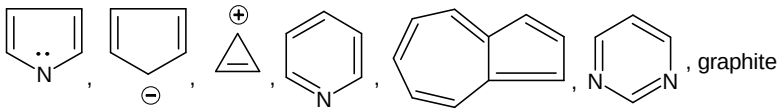
$SO_3^{2-} + HCl \longrightarrow H_2O + 2Cl^- + SO_2 \uparrow$

$HCO_3^- + HCl \longrightarrow H_2O + CO_2 \uparrow + Cl^-$

3.



5. Aromatic compounds



A layer of graphite (graphene) contains infinite lattice of fused aromatic rings. All the valencies are satisfied (except at the edges) and no bonds are needed between layers. (Planar layered structure of fused aromatic rings).

6. $5M \rightarrow 5 \text{ mol } H_2SO_4 = 5 \times 98 = 490 \text{ gm of } H_2SO_4$

$5M \rightarrow 1000 \text{ ml of solution} = 1000 \times 1.115 = 1115 \text{ gm solution}$

$\therefore$ wt. of solvent = 1115 – 490 = 625 gm

$$m = \frac{\text{moles of solute}}{\text{wt. of solvent}(\text{in kg})} = \frac{5}{625 \times 10^3} = 8 \text{ mol/kg}$$

7. $_{15}P \longrightarrow 1s^2, 2s^2, 2p^6, 3s^2 3p^3$

For 's' $\ell = 0$, m = 0, $e^- = 6$

For 'p', $\ell = 1$, m = -1, +1, 0



$= 2e^-$



$= 1e^-$

Total $e^- = 6 + 2 + 1 = 9$

8. $\underset{a}{PCl_5} \rightleftharpoons \underset{0}{PCl_3} + \underset{0}{Cl_2}$

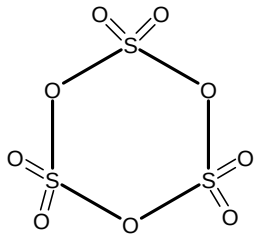
$\underset{a-ax}{\quad} \quad \underset{a\alpha}{\quad} \quad \underset{a\alpha}{\quad} \quad \alpha = 0.50$

$$K_p = \frac{\alpha^2 P}{1-\alpha^2} \Rightarrow \frac{K_p}{P} = \frac{0.5 \times 0.5}{0.75}$$

$$\Rightarrow \frac{P}{K_P} = 3$$

$\therefore P = 3 \times K_P$

9.

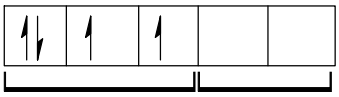


10. $\left[Mn(CN)_6\right]^{3-}$

$x + 6(-1) = -3$

$\therefore x = +3$

$Mn^{3+} \longrightarrow [Ar]3d^4 4S^0$



As CN^- is strong ligand hence due to splitting it will from low spin complex.

Mathematics

PART – III

SECTION – A

1. $4 = x_1 + \frac{1}{x_2} \geq 2\sqrt{\frac{x_1}{x_2}}$

$1 = x_2 + \frac{1}{x_3} \geq 2\sqrt{\frac{x_2}{x_3}}$

$4 = x_3 + \frac{1}{x_4} \geq 2\sqrt{\frac{x_3}{x_4}}$

$1 = x_4 + \frac{1}{x_1} \geq 2\sqrt{\frac{x_4}{x_1}} \Rightarrow 16 = \left(x_1 + \frac{1}{x_2}\right)\left(x_2 + \frac{1}{x_3}\right)\left(x_3 + \frac{1}{x_4}\right)\left(x_4 + \frac{1}{x_1}\right) \geq 2^4$

2. Any point on xy = 2 is $P\left(\sqrt{2}t, \frac{\sqrt{2}}{t}\right)$

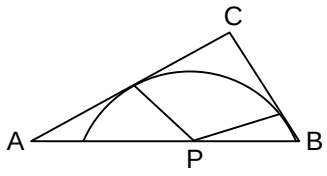
Normal at P is $\left(y - \frac{\sqrt{2}}{t}\right) = t^2\left(x - \sqrt{2}t\right)$

3. $\Delta_{ABC} = \Delta_{BPC} + \Delta_{APC}$

$\Delta = \frac{1}{2}ar + \frac{1}{2}br$

$r = \frac{2\Delta}{a+b}$

$\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2} = \sqrt{\frac{s(s-a)}{bc}} \cdot \sqrt{\frac{s(s-b)}{ac}} \cdot \sqrt{\frac{s(s-c)}{ab}} = \frac{s\Delta}{abc}$



4. $P(A_1 \cap A_2) = \frac{1}{4}$, $P(A_1) = \frac{2}{4}$

$P(A_2) = \frac{1}{2}$. Thus $P(A_1 \cap A_2) = P(A_1)\,P(A_2)$

5. $\frac{z-4}{z-2i} + \frac{\bar{z}-4}{\bar{z}+2i} = 0$ which is a circle $x^2 + y^2 - 4x - 2y = 0$

$xx_1 + yy_1 - 2(x + x_1) - (y + y_1) = x_1^2 + y_1^2 - 4x_1 - 2y_1$

It passes through (0, 0), so, the locus of (x_1, y_1) is $x^2 + y^2 - 2x - y = 0$.

So, $z_1 = x_1 + iy_1$, lies on this circle for which the points (2, 0) and (0, 1) are extremities of a diameter. Also (0, 0) and (2, 1) represent extremities of another diameter

6. $I = \int_{-\alpha}^{\alpha} e^x dx + \int_{-\alpha}^{\alpha} \cos x \ln\left(x + \sqrt{1+x^2}\right) dx$

$\Rightarrow e^{\alpha} - e^{-\alpha} > \frac{3}{2}$

7. $f(t) = t^3 - t^2 + t + 1$

$f'(t) > 0$

$\therefore$ max.$\{f(t): 0 \leq t \leq x\} = f(x)$

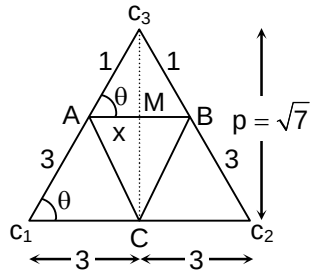
8. Put $2^{\left(2\sin^2 x-3\sin x+1\right)} = t$

9. $E_n = 3^n + {}^nC_2\cdot 3^{n-2} + {}^nC_4\cdot 3^{n-4} + \ldots..$
$O_n = {}^nC_1\cdot 3^{n-1} + {}^nC_3\cdot 3^{n-3} + \ldots..$
Adding and subtracting, we get $E_n + O_n = (3+1)^n = 4^n$, $E_n - O_n = (3-1)^2 = 2^n$
Adding and subtracting again, we get $2E_n = 4^n + 2^n \Rightarrow E_n = \dfrac{4^n + 2^n}{2}$
$2O_n = 4^n - 2^n \Rightarrow O_n = \dfrac{4^n - 2^n}{2}$

10. $\dfrac{dx}{dy} = x^2y^3 + xy$
$\dfrac{dx}{dy} - xy = x^2y^3$
$-\dfrac{1}{x^2}\dfrac{dx}{dy} + \dfrac{y}{x} = -y^3$
Put $\dfrac{1}{x} = t$

SECTION – C

1. $\cos\theta = \dfrac{3}{4} = \dfrac{x}{1}$
$\Rightarrow x = \dfrac{3}{4} \quad \Rightarrow AB = \dfrac{3}{2}$
$p = \sqrt{4^2 - 3^2} = \sqrt{7}$ and $c_3M = \dfrac{\sqrt{7}}{4}$
$\Rightarrow CM = \sqrt{7} - \dfrac{\sqrt{7}}{4} = \dfrac{3\sqrt{7}}{4}$



2. $x^2 + 1 = (x+i)(x-i)$
$b = 1$, $a = c$
Number of ways of choosing a, b, c = $10 = 10 \times 1$
$\therefore k = 1$

3. Unit vector $(\hat{n})$ perpendicular to $\overrightarrow{OA}$ and $\overrightarrow{CB}$
$\hat{n} = \dfrac{\overrightarrow{CB}\times\overrightarrow{OA}}{\left|\overrightarrow{CB}\times\overrightarrow{OA}\right|} = \dfrac{(3\alpha-7)\hat{i} - 4\hat{j} + (5-\alpha)\hat{k}}{\sqrt{(3\alpha-7)^2 + 16 + (5-\alpha)^2}}$
As $\overrightarrow{CB}\times\overrightarrow{OA} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha-3 & -1 \\ 1 & 2 & 3 \end{vmatrix} = \left|\overrightarrow{BA}\cdot\hat{n}\right| = \left|\dfrac{(7-3\alpha) - 4(2-\alpha) + 2(5-\alpha)}{\sqrt{(3\alpha-7)^2 + 16 + (5+\alpha)^2}}\right|$
Shortest distance = $\left|\overrightarrow{BA}\cdot\vec{n}\right|$

4. Replacing x by $\dfrac{1}{(1-x)}$, we obtain $f\left(\dfrac{1}{1-x}\right) + f\left(1-\dfrac{1}{x}\right) = -2x + \dfrac{2}{x}$
Again, replacing x by $1-\dfrac{1}{x}$

5. We have $a = \dfrac{x\left(x^6+1\right)}{x^2+1}$

$\Rightarrow x^6 + 1 = a\left(\dfrac{x^2+1}{x}\right) \geq 2a$

$\Rightarrow 2a - x^6 \leq 1$

6. $\Rightarrow (x^2 - 1)^2 = -3\{1 + \cos(ax^2 + bx + c)\}$ solution exists when L.H.S. = R.H. S. = 0
$\Rightarrow x = \pm 1$ and $\cos(ax^2 + bx + c) = -1$
When $x = 1, x = -1$
$\Rightarrow a + b + c = 3\pi, a - b + c = \pi$
$\Rightarrow b = \pi$ and $a + c = 2\pi$

7. $f(x) = \prod_{i=1}^{100} (x-i)^{i(101-i)}$

$\ln(f(x)) = \sum_{i=1}^{100} i(101-i)\ln(x-i)$

$\Rightarrow \dfrac{1}{f(x)} \cdot f'(x) = \sum_{i=1}^{100} \dfrac{i(101-i)}{(x-i)}$

8. For any quadrilateral,
$a + b + c > d$.
Now, $(a - b)^2 + (b - c)^2 + (c - a)^2 \geq 0$
$\Rightarrow a^2 + b^2 + c^2 \geq (ab + bc + ca)$
$\Rightarrow 3(a^2 + b^2 + c^2) \geq 3(ab + bc + ca)$
$\Rightarrow 3(a^2 + b^2 + c^2) \geq a^2 + b^2 + c^2 + 2(ab + bc + ca)$
$\Rightarrow 3(a^2 + b^2 + c^2) \geq (a + b + c)^2$
$\Rightarrow 3(a^2 + b^2 + c^2) > d^2$
$\Rightarrow \dfrac{(a^2+b^2+c^2)}{d^2} > \dfrac{1}{3}$.

$\therefore$ The minimum value of $\left(\dfrac{a^2+b^2+c^2}{d^2}\right)$ is $\dfrac{1}{3}$

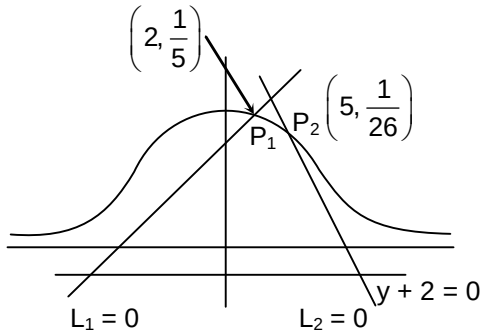
9. As $P(\alpha, (1 + \alpha^2)^{-1})$ lie on $y = \dfrac{1}{1+x^2}$

$\therefore$ On solving $y = \dfrac{1}{x^2+1}$ with L_1, we get

$\left(2, \dfrac{1}{5}\right)$ ..... (1)

and with line L_2, we get $P_2\left(5, \dfrac{1}{26}\right)$ ..... (2)

$\therefore$ Equation (1) and (2), we get $2 \leq \alpha \leq 5$



10. $f'(x) = \cos x - 2a \cos 2x - \cos 3x + 2a \geq 0 \ \forall \ x \in R$
$\Rightarrow 2 \sin^2 x (\cos x + a) \geq 0$
$\Rightarrow a \geq -\cos x \geq 1$